

4(a)	<table><tr><th>correct order</th><th>letter of stage</th></tr><tr><td>1</td><td>E</td></tr><tr><td>2</td><td>I</td></tr><tr><td>3</td><td>F</td></tr><tr><td>4</td><td>A</td></tr><tr><td>5</td><td>G</td></tr><tr><td>6</td><td>J</td></tr><tr><td>7</td><td>B</td></tr><tr><td>8</td><td>K</td></tr><tr><td>9</td><td>C/H</td></tr><tr><td>10</td><td>H/C</td></tr><tr><td>11</td><td>D</td></tr></table>	correct order	letter of stage	1	E	2	I	3	F	4	A	5	G	6	J	7	B	8	K	9	C/H	10	H/C	11	D	4
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; ; ; ;																										
4(b)	any four from: 1 acts as, coenzyme / cofactor ; 2 to dehydrogenase / in dehydrogenation ; 3 takes / carries / delivers / releases / gives, hydrogen / H (atoms) ; 4 to, (carrier on) ETC / inner mitochondrial membrane / cristae or for, ATP production / oxidative phosphorylation ; 5 <i>ref. to</i> redox / reduction and oxidation / reduced and oxidised ; 6 2.5 / 3, ATP per reduced NAD ;	4																								

Question	Answer	Marks
4(c)(i)	6 ;	1
4(c)(ii)	<i>any one from:</i> 1 hydrogen carbonate (ions) prevents decrease in (blood) pH ; 2 buffer system / carbonic acid dissociates / <i>ref. to</i> carbonic anhydrase ;	1

Question	Answer	Marks
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